

Session 10:

Strengths & Limits of Generative AI

You've created with AI. You've seen it hallucinate. You've learned how it works under the hood. Now it's time to step back and see the big picture.

What is AI exceptionally good at? What should you avoid using it for? And how do you decide when to trust it?

Today we answer these questions!



Objectives

By the end of this session, you will be able to:

01

Identify common strengths of Generative AI

02

Identify common limitations of Generative AI

03

Decide when to use AI and when NOT to use it

04

Apply the Human-in-the-Loop approach to responsible AI use

05

Analysis of a Generative AI tool

Recap - What We Learned in Session 9

AI Hallucination	When AI confidently generates false information
Confidence Score	A percentage showing how sure the AI seems about its answer
Accuracy	Whether the information is actually correct
Why Hallucinations Happen	No real understanding + pattern matching gone wrong + data gaps
Famous Examples	Lawyer who used fake ChatGPT cases; Airline chatbot gave wrong policy
Types of Hallucinations	Factual, Source, Logical, Entity, Relationship
How to Spot	Too specific without sources, mixes real/fake, contradicts common knowledge
How to Reduce	Be specific, provide context, ask for sources, verify output
Tool Difference	NotebookLM refuses to answer without sources; ChatGPT always tries to answer

📌 **Remember:** AI can be confidently wrong. **Confidence ≠ accuracy**. Always verify important information.



What GenAI Does Well (Strengths)

Generative AI has several **common strengths** - things it's generally good at. Different AI tools may have different strengths, and you might discover more as you use them. Here are some of the most common ones:

Strengths:

Speed

AI generates content in seconds. A task that might take you an hour can be drafted in moments.

Example: Your school's environmental club needs a one-page flyer announcing their first meeting. You describe the event to AI, and within 10 seconds, you have three different draft versions with headings, bullet points, and a call to action.

Brainstorming

AI produces endless ideas when you're stuck.

Example: Your class is organizing a "Community Heroes Day" to thank local firefighters, nurses, and teachers. You need 20 creative ways to show appreciation. AI generates ideas like "handmade thank-you cards," "a video montage of student messages," "certificates of appreciation," and "a thank-you banner signed by every class."

Strengths:

Explaining Complex Ideas

AI can rephrase difficult concepts in simpler language.

Example: You're trying to understand how batteries work for a science project. You ask AI: *"Explain how a battery works using a water flow analogy."* AI responds: *"Think of a battery like a water pump. The pump pushes water through pipes - that's the current. The pressure pushing the water is like voltage. The width of the pipe is like resistance."*

Translation

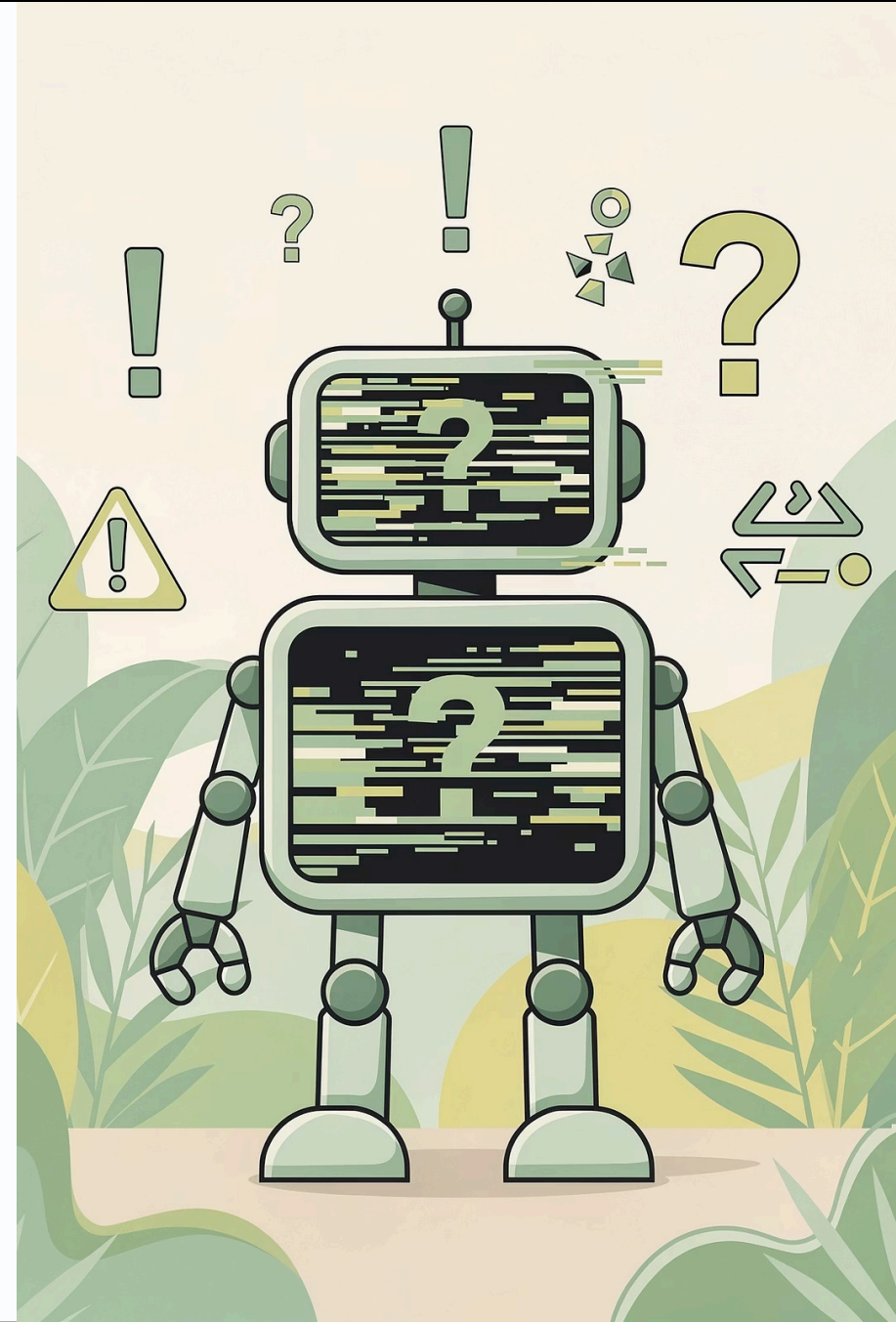
AI helps bridge language gaps

Example: A new student from Brazil just joined your class. They're still learning English. You want to write a note inviting them to sit with you at lunch. AI helps translate your message to Portuguese so they feel welcome on their first day.

- ❏ **Remember:** Generative AI has certain capabilities built into how it works. These are **common strengths** – things many AI tools tend to do reasonably well. But here's the important part: **not every AI tool excels at all of these**, and some tools have strengths you won't discover until you use them.

What GenAI Struggles With (Limits)

Generative AI also has **common limits** - things it generally struggles with. Different AI tools may have different weaknesses, and you'll likely encounter more as you use them. Here are some of the most important ones:



Limits :

No Real Understanding

AI mimics patterns but doesn't comprehend meaning.

Example: Ask AI: *"If you drop a feather and a bowling ball from the same height, which hits the ground first?"* AI might correctly say "bowling ball" but it doesn't understand gravity, air resistance, or physics. It's just repeating patterns from its training. It couldn't explain why this happens or what would change on the moon.

Outdated or Missing Information

AI only knows what was in its training data, which has a cutoff date.

Example: Your school just announced a new policy about bringing phones to class. You ask AI for details. AI says it doesn't have any information because the policy was created after its last update. You need to check the school's actual email or website.

Limits :

Hallucinations

AI confidently makes things up.

Example: You ask AI about the history of your town's oldest park. AI responds: *"Riverside Park was created in 1895 by famous landscape architect Frederick Olmsted..."* When you check with the local historical society, you discover the park was actually created in 1910, there's no connection to Olmsted, and the bridge was built in 1925. The AI invented an impressive-sounding but completely false history.

Bias

AI repeats stereotypes from its training data.

Example: You ask AI to generate a list of "famous athletes." Without specific prompting, it might list mostly male athletes from popular sports. It's not intentionally biased - it's reflecting patterns in the data it was trained on, which historically gave more media coverage to male athletes. You would need to specifically ask for "famous female athletes" or "athletes from different countries" to get a more complete picture.

❏ **Remember:** These are common limits, but not every AI tool struggles with all of them. Part of becoming a smart AI user is learning the specific strengths and limits of the tools you use.

Mini Activity 1 - Strength or Limit?

For each scenario, decide: Is this a **STRENGTH** of AI or a **LIMIT**?

#	Scenario	Strength or Limit?
1	You need to create a sign-up sheet for your class picnic	—
2	You ask AI what time the library closes today, and it gives you hours from last year	—
3	You're writing a thank-you card for a teacher and need help with the opening line	—
4	You ask AI for a joke to tell at a family dinner, and it gives you a complex physics joke	—
5	You're curious about why onions make you cry and want a simple explanation	—
6	AI generates a list of "famous artists" that includes only European painters	—
7	You need a packing list for a weekend trip to the beach	—
8	AI tells you that a new movie starring your favorite actor came out last week, but when you search, no such movie exists	—

Mini Activity 1 - Answer Key

#	Scenario	Answer	Why?
1	Class picnic sign-up sheet	Strength	Creating simple templates and forms is something AI can help with
2	Wrong library hours	Limit	AI may have outdated information - a common limit for time-sensitive facts
3	Stuck on thank-you card opening	Strength	Getting started with writing is something AI helps with
4	Physics joke	Limit	AI doesn't understand what's age-appropriate or funny for a family dinner
5	Why onions make you cry	Strength	Explaining everyday curiosities in simple terms is something AI does well
6	Only European artists	Limit	AI reflects bias in its training data - an important limit to watch for
7	Beach trip packing list	Strength	Drafting and organizing checklists is something AI can help with
8	Fake movie release	Limit	Classic hallucination - AI sounds confident but made up something that doesn't exist

When to Use AI

You might use AI when your goal is to explore, generate ideas, or create something new and you plan to review and personalize the output yourself.

- ❏ **Example:** Your class is putting together a "Welcome Back" packet for new students. You use AI to draft a list of frequently asked questions about the school. Then you review each question, add missing ones, and verify all answers with the school handbook. **AI helped you start; you finished with accuracy.**



When to Use AI - Situations & Examples

- ① The situations below are **general guidelines, not rigid rules**. There may be times when you don't use AI even in these situations, and that's perfectly fine. There may also be times when you use AI for things not on this list. The goal isn't to create a checklist of when you *must* use AI, it's to help you recognize opportunities where AI might genuinely support your thinking, not replace it.

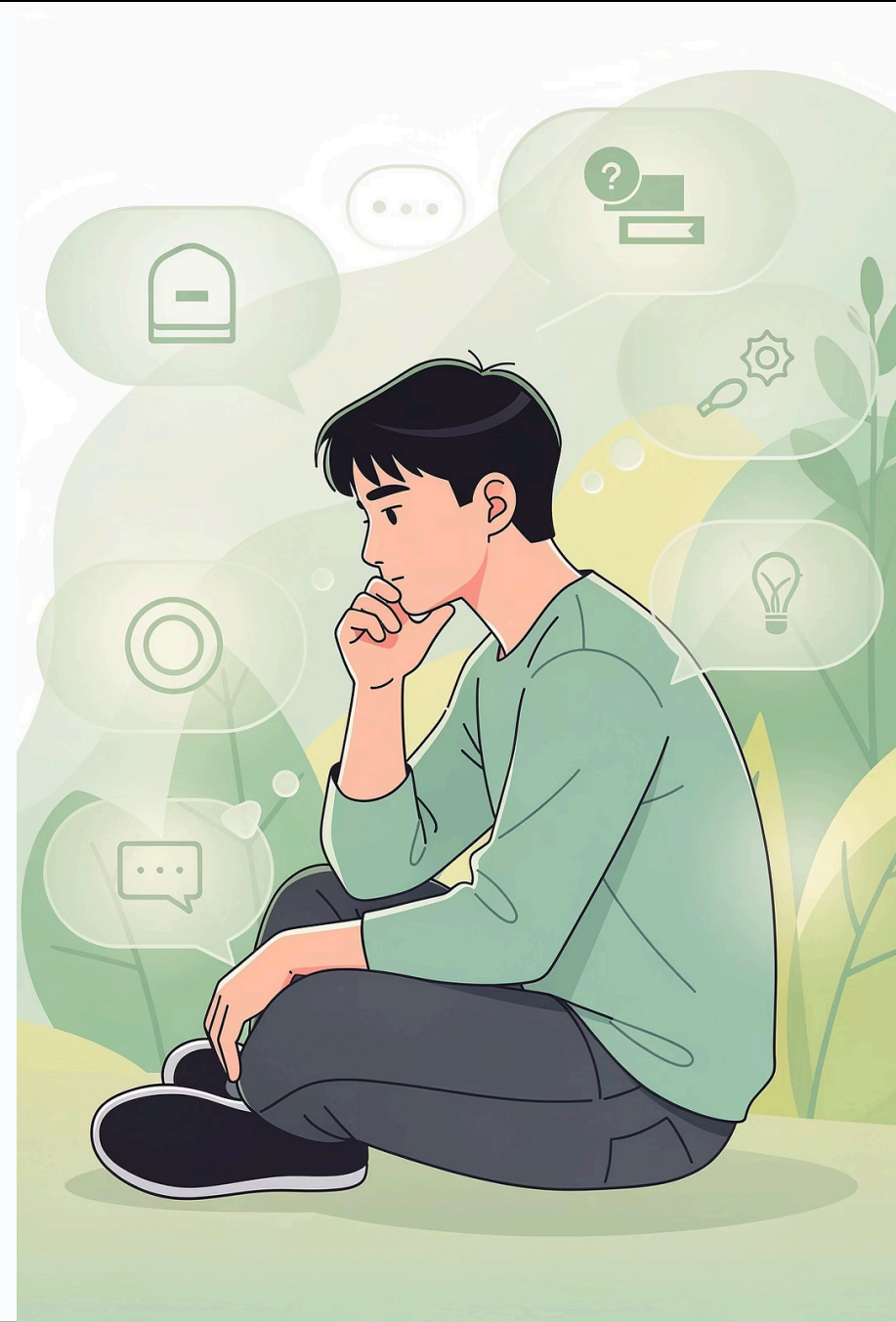
Situation	Example	Explanation
Exploring creative angles	"Give me three different ways to announce our school talent show - one funny, one exciting, one formal"	You see different approaches and choose what fits your audience
Building upon a partial idea	"I have an idea for a story about a time-traveling backpack. Help me expand it into a plot outline."	You provide the core idea; AI helps you develop possibilities

Simplifying instructions	"Make this recipe easier to understand for someone who's never baked before"	AI rephrases complex steps; you still need to actually follow them
Finding blind spots	"Here's my packing list for a hiking trip. What might I have forgotten?"	AI suggests common items; you decide what's relevant
Translating intent, not just words	"I want to politely decline an invitation without hurting feelings. Help me phrase this."	AI offers options; you know the person, so you choose what feels right
Creating variations	"Take this announcement and make it shorter for social media"	AI adapts your content; you review and approve
Comparing perspectives	"Show me arguments for and against having school uniforms"	AI presents multiple viewpoints; you form your own opinion

When NOT to Use AI

DO NOT use AI when accuracy is critical or harm could result from mistakes.

- ❏ **Example:** Your friend has a headache and asks AI: "What medicine should I take?" AI suggests a random painkiller without knowing their allergies, age, or medical history. This is dangerous. Instead, they should ask a doctor.



When NOT to Use AI - Situations & Examples

- ① The situations below are **general guidelines, not rigid rules**. Think of them as "yellow flags", reasons to pause and think carefully before using AI. There may be times when you use AI in these situations anyway, and that's okay as long as you understand the risks and take responsibility for verifying the output. The goal isn't to never use AI, it's to be thoughtful about when you do.

Situation	Example	Explanation
You need accurate, current facts	"What time does camp registration close?"	AI may give wrong info; check official sources
You're making a personal decision	"Should I apologize to my friend?"	AI doesn't know your friendship; trust your own feelings

Someone needs emotional support	"What should I say to someone who is grieving?"	AI misses human complexity; talk to a trusted person
You're doing important school work	Submitting AI-generated work as your own	AI's output comes from others' work
You'd be responsible for the outcome	Telling a friend to follow AI's advice	If AI is wrong, you're responsible
You can't check if it's correct	Using AI's output without verification	You're trusting something that might be wrong
You're sharing private information	Entering your address or passwords	Your info may not be private; keep personal details to yourself
Health or safety is involved	"Is this berry safe to eat?"	AI could make up dangerous advice; ask a knowledgeable adult

Mini Activity 2 - Use AI or Avoid AI?

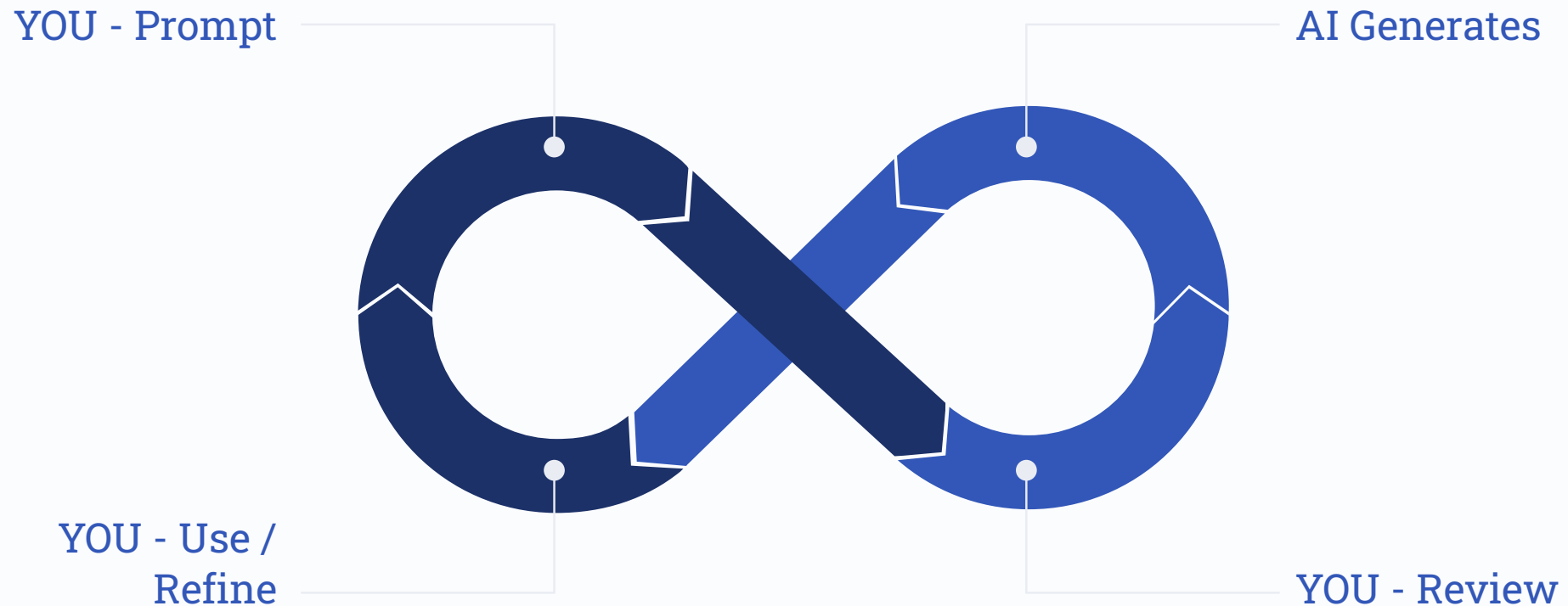
#	Scenario	Use or Avoid?
1	You need to find out when early registration ends for summer sports camp	
2	You're trying to come up with a name for your team	
3	Your class is selling used books and you need suggested price ranges	
4	You woke up with a fever and aren't sure if you should go to school	
5	You're writing a speech for a school event and need help	
6	You need to remember what time your group project meeting is tomorrow	
7	You're looking for local places that might accept teen volunteers on weekends	
8	You're trying to decide whether to confront someone about something they said, but you're not sure if you misinterpreted it	

Mini Activity 2 - Answer Key

#	Scenario	Answer	Why?
1	Summer camp registration deadline	AVOID	AI may have old dates - check the camp's website or email
2	Team name	USE	AI can generate name ideas - your team picks what fits your group
3	Used book price ranges	USE	AI can suggest typical prices - you adjust based on book condition
4	Fever, should you go to school	AVOID	AI doesn't know your symptoms - ask a parent or guardian
5	Opening lines for a speech	USE	AI can draft an opening - you add your own topic and examples
6	Group project meeting time	AVOID	AI doesn't have your schedule - check your calendar or group chat
7	Weekend teen volunteer opportunities	USE	AI can suggest places to check - you research which ones are currently accepting
8	Confront someone about something they said	AVOID	AI doesn't know the situation or your relationship - talk to a trusted friend or adult first

The Human-in-the-Loop Approach

A framework where **AI makes suggestions, but humans make the final decisions**. The human is always "in the loop" - involved, checking, and responsible.



Human-in-the-Loop: Planning a School Assembly

Here's how the loop works in a real example:

Step	Who	Action	What Happens
1	YOU	Prompt	"Give me ideas for a school assembly about kindness"
2	AI	Generates	Suggests 10 ideas: guest speakers, student performances, videos, etc.
3	YOU	Review	"These are good starting points, but we don't have budget for guest speakers. Let's focus on student performances."
4	YOU	Refine	New prompt: "Now give me 5 ideas for student performances about kindness"
5	AI	Generates	Suggests skits, songs, poetry readings, art displays, etc.
6	YOU	Decide	"The skit and poetry reading ideas are perfect. Let's form committees."

 **YOU were in the loop the entire time.** AI assisted; you directed, reviewed, and decided.



The Pilot and Autopilot

Pilots don't just turn on autopilot and go to sleep. They:

Program the autopilot with the correct route

Monitor its decisions constantly

Take over immediately if something seems wrong

Are always responsible for the flight

☐ **AI is your autopilot. YOU are the pilot. You're always in charge.**

Hands-On Activity - Generative AI Analysis

Today you'll write up your analysis of ChatGPT, Gemini, or NotebookLM - capturing what you've learned about how it works, where it struggles, and when to use it thoughtfully.

#	Section	What to Write
1	Tool You Analyzed	Which tool? Why did you choose it?
2	What It Does Well	2–3 strengths you observed (with examples from your sessions)
3	What It Struggles With	2–3 limits you noticed (with examples)
4	Your Best Prompt	One prompt that worked really well - and what it created
5	Hallucination Example	One time it was wrong - what it said, what was correct, how you verified
6	When to Use This Tool	Situations where you WOULD use this tool
7	When to Avoid This Tool	Situations where you would NOT use this tool
8	Your Personal Rule	Based on Human-in-the-Loop, what's your rule for using AI?

Use this template to organize your Generative AI Analysis:

Tool I Analyzed Tool I Analyzed: 	What It Does Well 1. Example: 2. Example: 	What It Struggles With 1. Example: 2. Example: 	My Best Prompt Prompt I used: What it created: Why this prompt worked well:
Hallucination Example What I asked: What AI said: What was actually correct: How I verified this:		When I WOULD / Would AVOID Use This Tool Would Use: • • • Would Avoid: • • •	



My Personal Rule for Using AI (based on Human-in-the-Loop):

Summary

Concept	Key Idea
Common Strengths	Speed, brainstorming, drafting, explaining, translating and more!
Common Limits	No understanding, outdated info, hallucinations, no common sense, bias and more!
When to Use AI	Brainstorming, drafting, getting unstuck, generating options
When to Avoid AI	Facts that matter, health/safety, personal info, important decisions
Human-in-the-Loop	AI assists - YOU decide, review, and are responsible
Your Analysis Project	Your complete analysis of a Generative AI tool

📌 **Most Important Takeaway:** AI is a powerful tool with many strengths and limits. Different AI tools have different strengths and limits. **YOUR job is to discover them, use them wisely, and always be the human in the loop.**

Homework - Part 1: Quick Check

Match each term to its meaning:

#	Term	Letter	Meaning
1	Strength	A.	Situations where AI is risky
2	Limit	B.	What AI is generally good at
3	Oversight	C.	What AI generally struggles with
4	Use AI	D.	Using AI where it's helpful, not harmful
5	Avoid AI	E.	Human supervision of AI outputs
6	Appropriate Use	F.	Situations where AI is helpful

Answers: 1-___ , 2-___ , 3-___ , 4-___ , 5-___ , 6-___

Homework - Part 2: Apply What You Learned

Find **one example in your daily life this week** where you could use AI appropriately (**USE AI**) and one example where you should NOT use AI (**AVOID AI**).

	Your Example	Why
USE AI		
AVOID AI		

Next Session Preview

Session 11: The Art of Prompting

Now that you know AI's strengths and limits, it's time to get better at talking to it.

In Session 11, you'll learn:

What prompt is

Why vague instructions get bad results

How to add context and constraints

How to refine prompts to get what you need